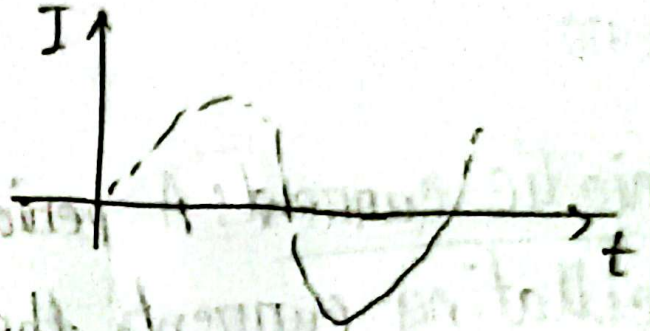
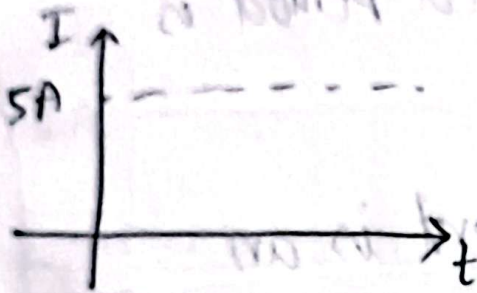
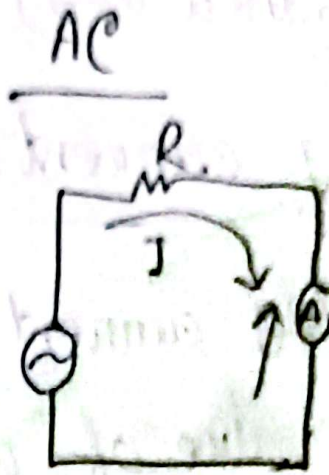
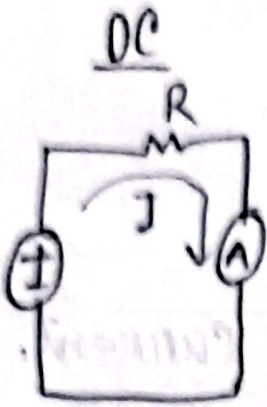
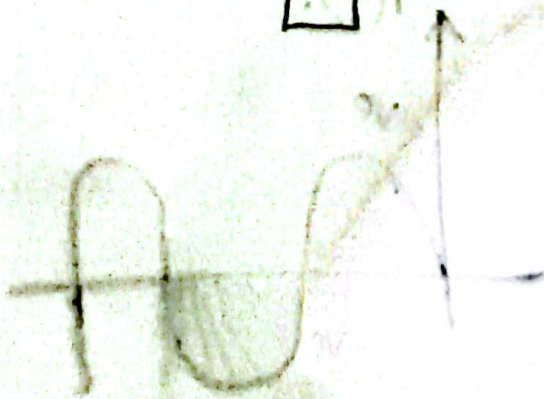
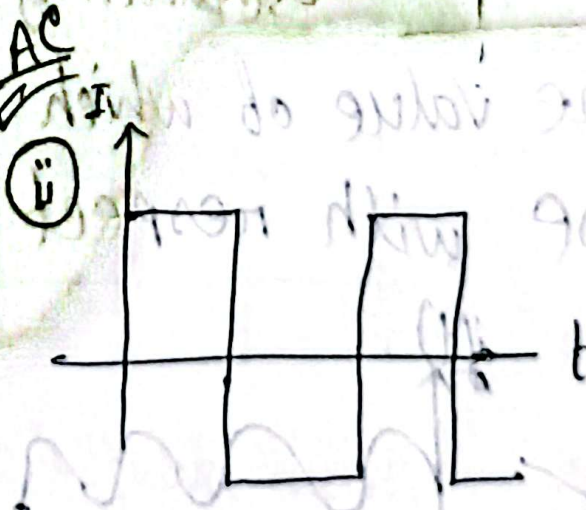
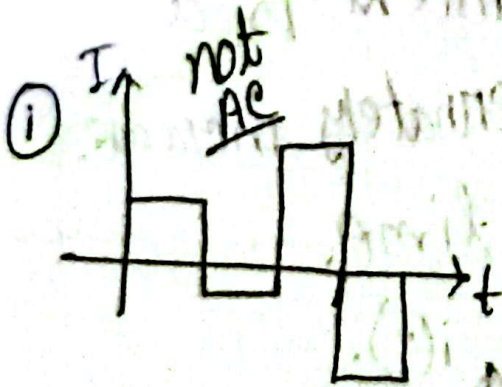




- (i) Direction
- (ii) Periodicity
- (iii) Average = 0

for
AC



AC

not AC
(average value
not zero)

oscillating
current

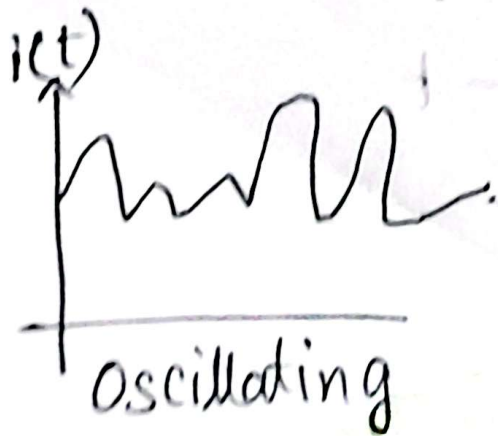
Class-1 (Sadiku e-2)

1) Alternating current (AC)

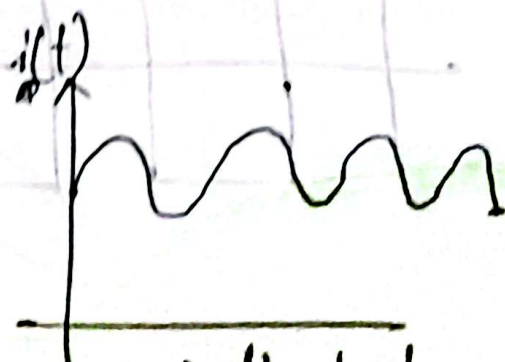
An alternating current is a periodic current, the average value of which over a period is zero.

Periodic current: A periodic current is an oscillating current, the value of which recur (repeat) at equal interval of time.

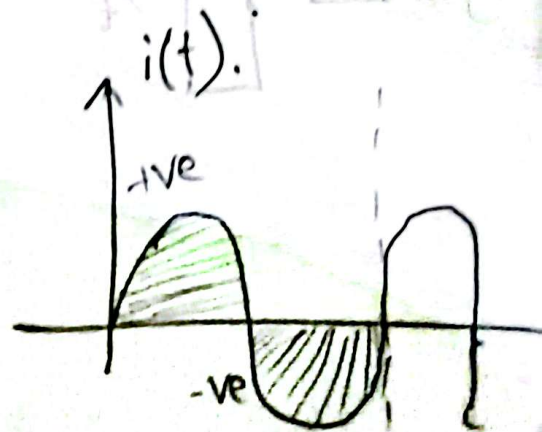
Oscillating current: Oscillating current is a current the value of which alternately increase and decrease with respect to time.



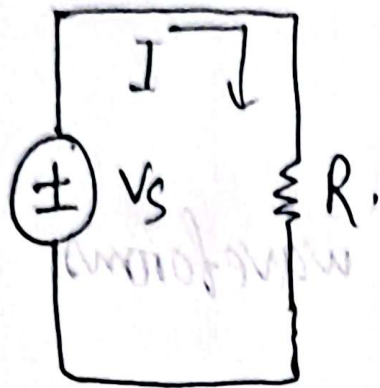
Oscillating
Current



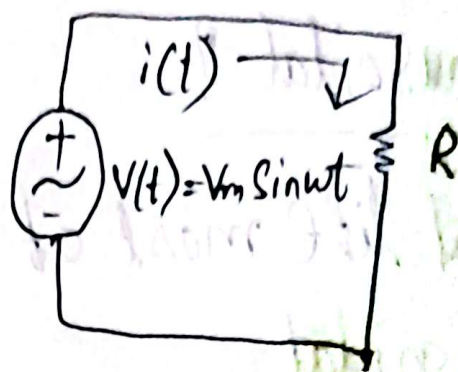
Periodic but
not AC
(average value
not zero)



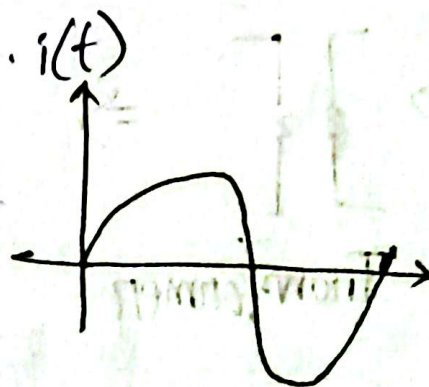
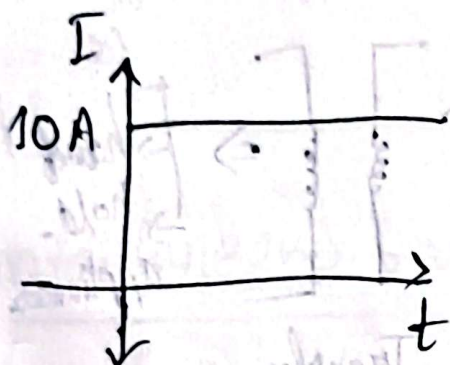
AC Signal.



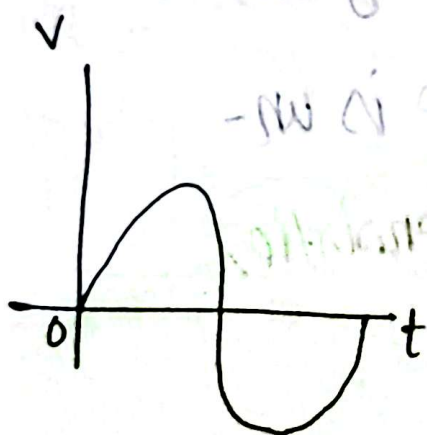
a) DC source



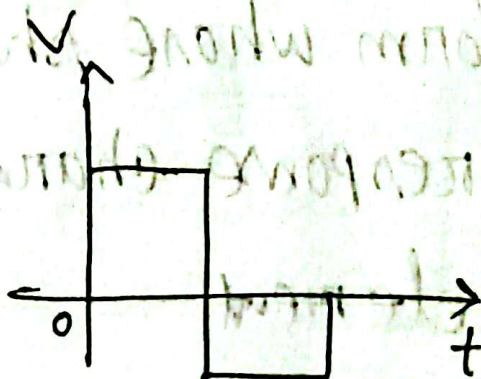
b) AC source



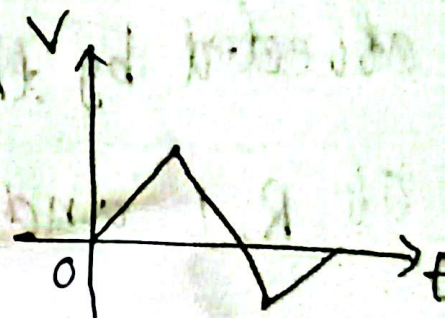
Examples of different AC waveforms



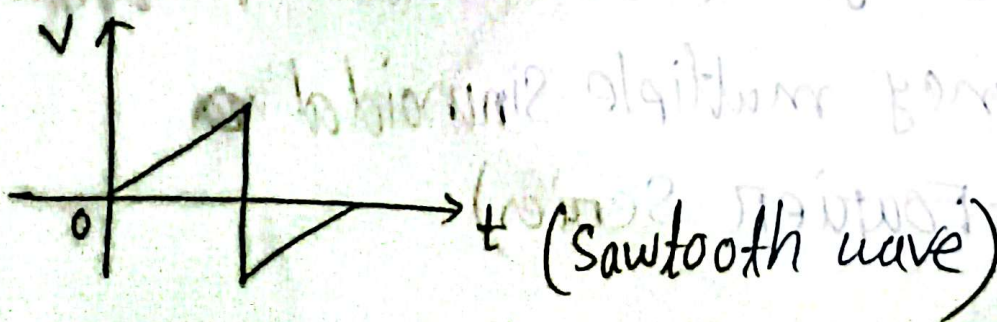
a) Sinusoidal
(Sine and cosine)



b) Square wave



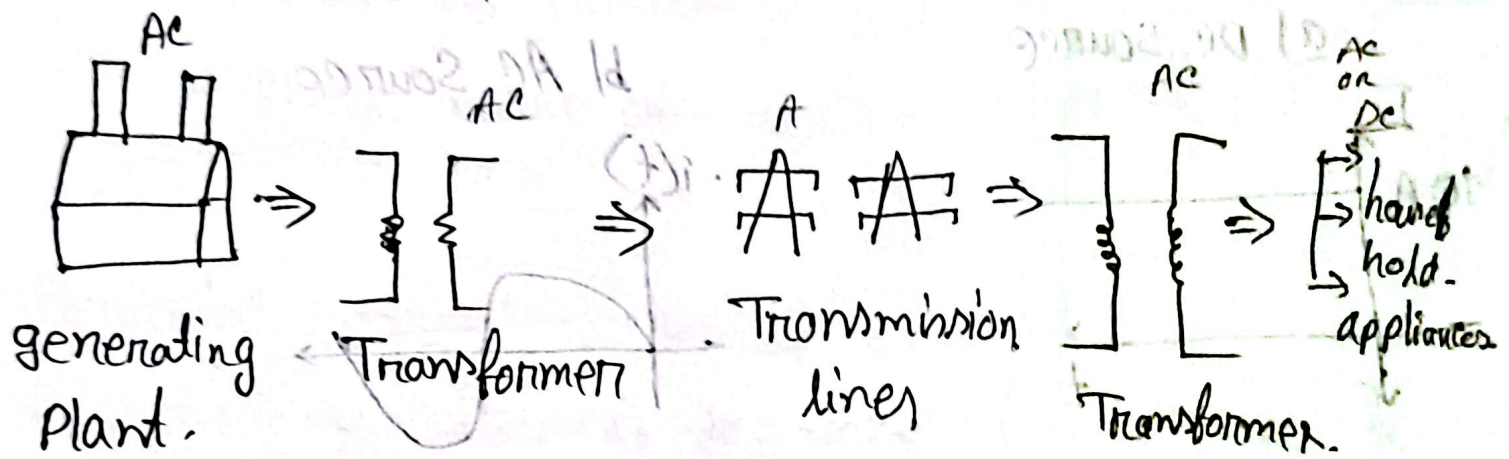
c) Triangular wave



(Sawtooth wave)

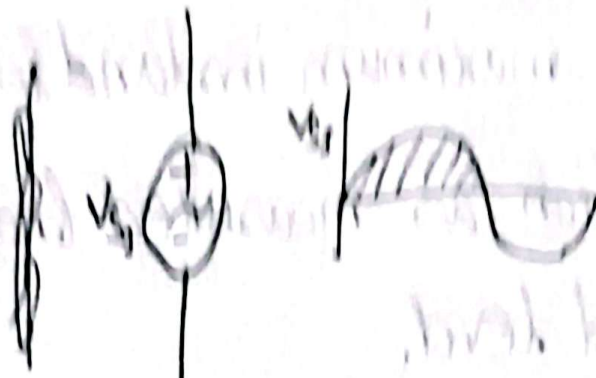
why sinusoidal?

In real life most of the AC waveforms are sinusoidal.

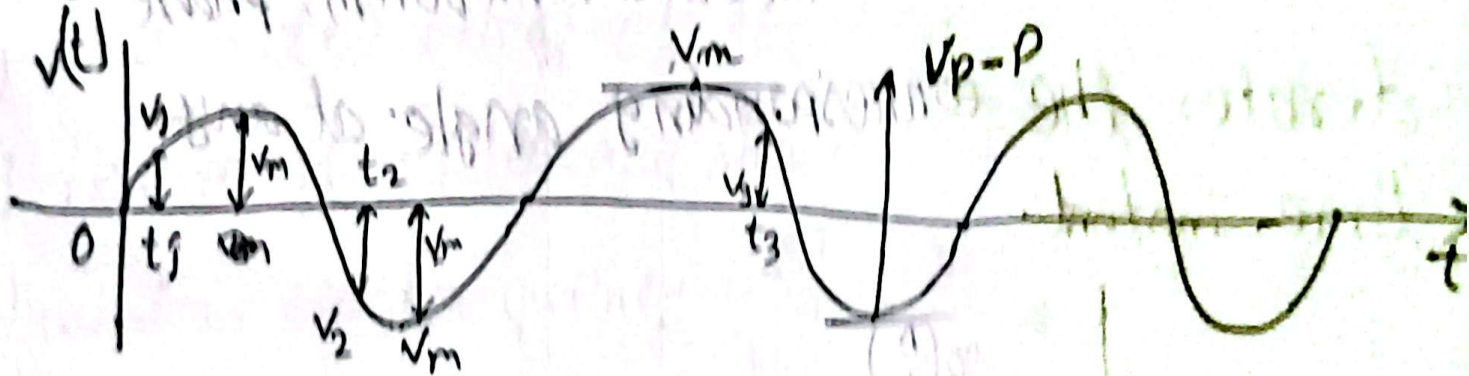


The sinusoidal waveform is the only alternating ~~and~~ waveform whose shape is unaffected by the response characteristics of R, L and C element.

Theoretically, any periodic waveform can be represented by the summation of different frequency multiple sinusoidal waveform (Fourier Series)



Instantaneous value: The magnitude of a wave form at any instant. Here v_1, v_2, v_3 are instantaneous values at time t_1, t_2 and t_3 respectively.

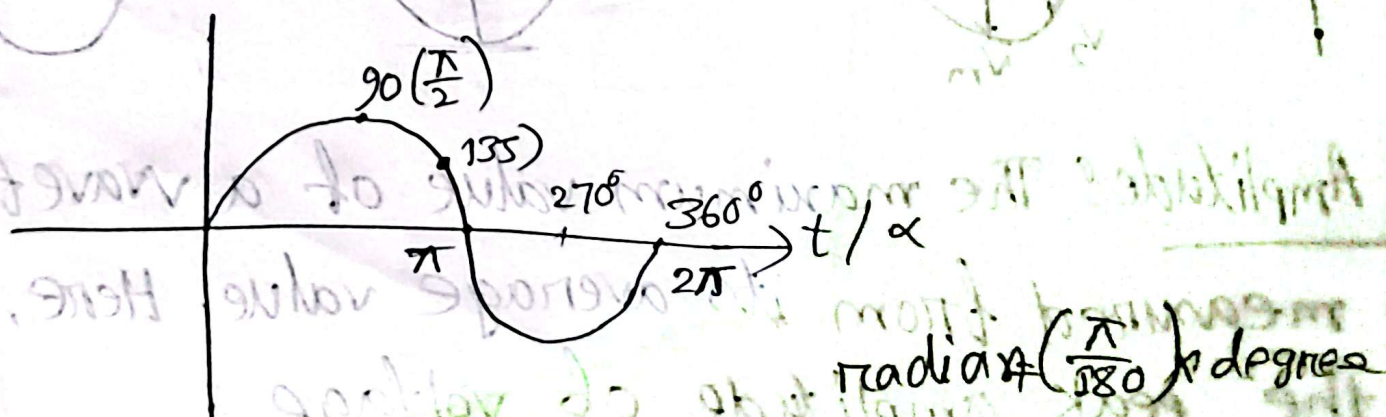


Peak Amplitude: The maximum value of a waveform as measured from its average value. Here, V_m is the peak amplitude of voltage.

Peak value: The maximum instantaneous value of a function as measured from the zero to r.m.s. level.

Peak to peak value: It represents the full voltage between positive and negative peaks of the waveform. V_{p-p} denotes the peak to peak value.

Phase: For a sinusoidal waveform, phase denotes the corresponding angle at any time instant.



Angular Frequency = $\frac{\text{Angular displacement}}{\text{time}}$.

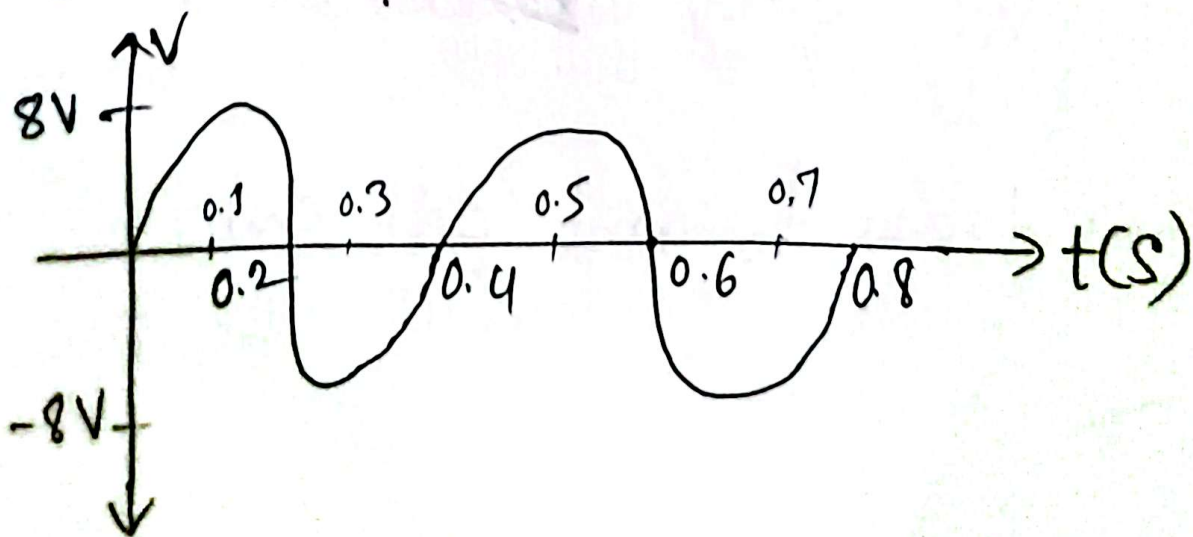
$$\omega = \frac{\alpha}{t} \text{ (rads}^{-1}\text{)}$$

$$\omega = \frac{2\pi}{T} \text{ [Time period = } T\text{]}$$

$$\boxed{\omega = 2\pi f} \text{ [} f = \frac{1}{T}\text{]}$$

Q) For the following sinusoidal waveform:

- what is the peak value?
- what is the instantaneous value at 0.3 s and 0.6 s?
- what is the peak to peak value of the ~~waveform~~ waveform?
- what is the period of the waveform?
- How many cycles are shown?
- what is the frequency and angular frequency of the waveform?



Ans

a) The peak value is 8V

b) 0.3s $\rightarrow$ -8V

0.6s $\rightarrow$ 0V

c) maximum-minimum
 $= 8 - (-8)$

$$= 16V$$

d) $T = 0.4s$

e) 2 cycles.

$$f) f = \frac{1}{T} = \frac{1}{0.4} = 2.5 \text{ Hz}$$

$$\omega = 2\pi f = 2\pi(2.5) \\ = 5\pi \text{ rad s}^{-1}$$

